

Supplemental Information

Supplemental Methods

Materials

Sodium acetate ($\geq 99\%$), sodium lactate ($\geq 99\%$), sodium pyruvate ($\geq 99\%$), glycerol ($\geq 99\%$), thymidine ($\geq 99\%$), 3-methyl-2-oxobutanoic acid (95%), 4-hydroxybenzoic acid ($\geq 99\%$), kynurenine ($\geq 98\%$), 3-methyl-2-oxopentanoic acid ($\geq 98\%$), 4-methyl-2-oxopentanoic acid ($\geq 98\%$), leucine ($\geq 99.5\%$), isoleucine ($\geq 99.5\%$), valine ($\geq 99.5\%$), ammonium chloride ($> 99.5\%$), sodium phosphate ($\geq 99.5\%$), EDTA ($\geq 99\%$), and iron (III) chloride ($\geq 98\%$) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Deionized water was obtained from a Milli-Q system (Millipore; resistivity $18.2\text{ M}\Omega$ at 24°C , $\text{TOC} < 1\text{ }\mu\text{M}$). Organic substrate and nutrient stocks were filtered through $0.2\text{ }\mu\text{m}$ nylon syringe filters (Fisher or Corning) under sterile conditions prior to addition to culture media. Cultures were maintained and uptake experiments were performed in 125 mL polycarbonate bottles, which were acid washed in 10% hydrochloric acid and autoclaved before use.

Growth experiment optimization

Prior to uptake experiments, experiments were performed to confirm *Alteromonas* growth on minimal media and lower-carbon media, and to determine the change in carbon concentration required to detect a statistically significant difference in culture carrying capacity. All optimization experiments contained the same inorganic nutrients as found in ProMM (Berube et al., 2015). Experiments were performed on black walled 96-well plates (Corning, Arizona) using a BioTek Synergy 2 plate reader, measuring OD_{600} every 30 minutes for 48 hours.

To test the ability of *Alteromonas* to grow on minimal media, we grew cultures under one of three conditions: ProMM (which contains pyruvate, acetate, lactate, and glycerol), pyruvate with vitamins, or pyruvate alone. For the “pyruvate” and “pyruvate with vitamins” conditions, pyruvate was added to 6.9 mM carbon, matching the total carbon concentration in ProMM. For the “pyruvate and vitamins” condition, vitamins were added at the same concentration as ProMM. Over the course of 48 hours, there was no significant difference in culture carrying capacity between conditions (ANOVA, $p > 0.05$, Figure S1).

To test the effect of growing *Alteromonas* on lower carbon concentrations than found in ProMM, we grew cultured on either 6.8, 5.5, 5, 4.1, 2.7, or 0 mM carbon from pyruvate. We found that *Alteromonas* could grow on lower carbon concentrations than those in ProMM (Figure S2), and that changing carbon concentrations by 0.9 mM C was sufficient for observing a significant difference in carrying capacity.

Flow cytometry

Samples for flow cytometry were stained for 10 minutes with SYBR Green 1 (1X final concentration) and diluted with a Tris-HCl/EDTA buffer to approximately 5×10^4 cells mL^{-1} prior to analysis. Cells were then quantified using a Guava easyCyte HT flow cytometer (Millipore) using a 488nm laser and 530nm emission filter. Instrument flow rate was set to medium ($0.59 \mu\text{L s}^{-1}$). *Alteromonas* was gated using green fluorescence and side scatter using an elliptical gate that encompassed the range of the population. All flow cytometry files were analyzed using guavaSoft InCyte 2.7.

Liquid chromatography-tandem mass spectrometry (LC-MS/MS)

Samples were analyzed in randomized order by ultra-performance liquid chromatography (UPLC, Accela Open Autosampler and Accela 1250 Pump, Thermo Scientific) coupled via heated electrospray ionization (H-ESI) to a triple quadrupole mass spectrometer (TSQ Vantage, Thermo Scientific) operated under selected reaction monitoring (SRM) mode. Spray voltage was set at 4000 V (positive mode) and 3200 V (negative mode). Source gases were set at 55 (sheath) and 20 (auxiliary), while the heated capillary temperature was 375 °C, and the vaporizer temperature was 400 °C. Chromatographic separation was performed on a Waters Acquity HSS T3 column (2.1 × 100 mm, 1.8 µm) equipped with a Vanguard pre-column and maintained at 40 °C. The column was eluted with (A) 0.1% formic acid in water and (B) 0.1% formic acid in acetonitrile at a flow rate of 0.5 mL min⁻¹. The gradient starts at 1% B for 1 min, ramps to 15% B from 1-3 min, ramps to 50% from 3-6 min, ramps to 95% B from 6-9 min, holds at 95% B until 10 min, ramps to 1% from 10-10.2 min, with re-equilibration at 1% B (total gradient time 12 min). Separate autosampler injections of 5 µL each were made for positive and negative ion modes. Pooled samples were run every five samples to act as a quality control. Raw files were converted to mzML using MSconvert (Chambers et al., 2012), peak areas were integrated in EL-MAVEN (Agrawal et al., 2019), curve fitting and quantification was performed in MATLAB, and statistical analysis was performed in R (R Core Team, 2021).

Search for 2-oxoacid transporters in the *Alteromonas macleodii* MIT1002 genome

We were unable to find annotated 2-oxoacid transporters in the *Alteromonas macleodii* MIT1002 genome (GenBank reference OX359241.1). Therefore, we searched for genes in the *A. macleodii* MIT1002 genome with homology to known prokaryote 2-oxoacid transporters. We

68 identified two such prokaryotic transporters using the transporter classification database (TCDB)
69 (TCDB; Saier et al., 2021), one in a photosynthetic purple nonsulfur bacterium, *Rhodobacter*
70 *capsulatus* SB1003 (Thomas et al., 2006), and the other in a nitrogen-fixing cyanobacteria,
71 *Anabaena* PCC7120 (Pernil et al., 2010). Each of the identified transporters is tripartite. We
72 performed a BLASTx (Altschul et al., 1990) search for each protein subunit against the
73 translated *A. macleodii* MIT1002 genome, and found one gene with low homology (30%
74 identity) to the large subunit of the *Anabaena* 2-oxoacid transporter protein (Supplemental Table
75 1). No other genes were identified in the *A. macleodii* MIT1002 genome with any homology to
76 known prokaryote 2-oxoacid transporter genes. While our search for 2-ketoacid transporters did
77 not offer new leads, characterization of transporters in other marine microbes has provided
78 deeper insight into labile dissolved organic matter cycling (Schroer et al., 2023). Further
79 characterization of *A. macleodii* transporters is thus a promising avenue for future research.

80 **Supplemental Tables**

81 Table S1. Prokaryotic 2-oxoacid transporter genes used to search for homologous *A. macleodii*

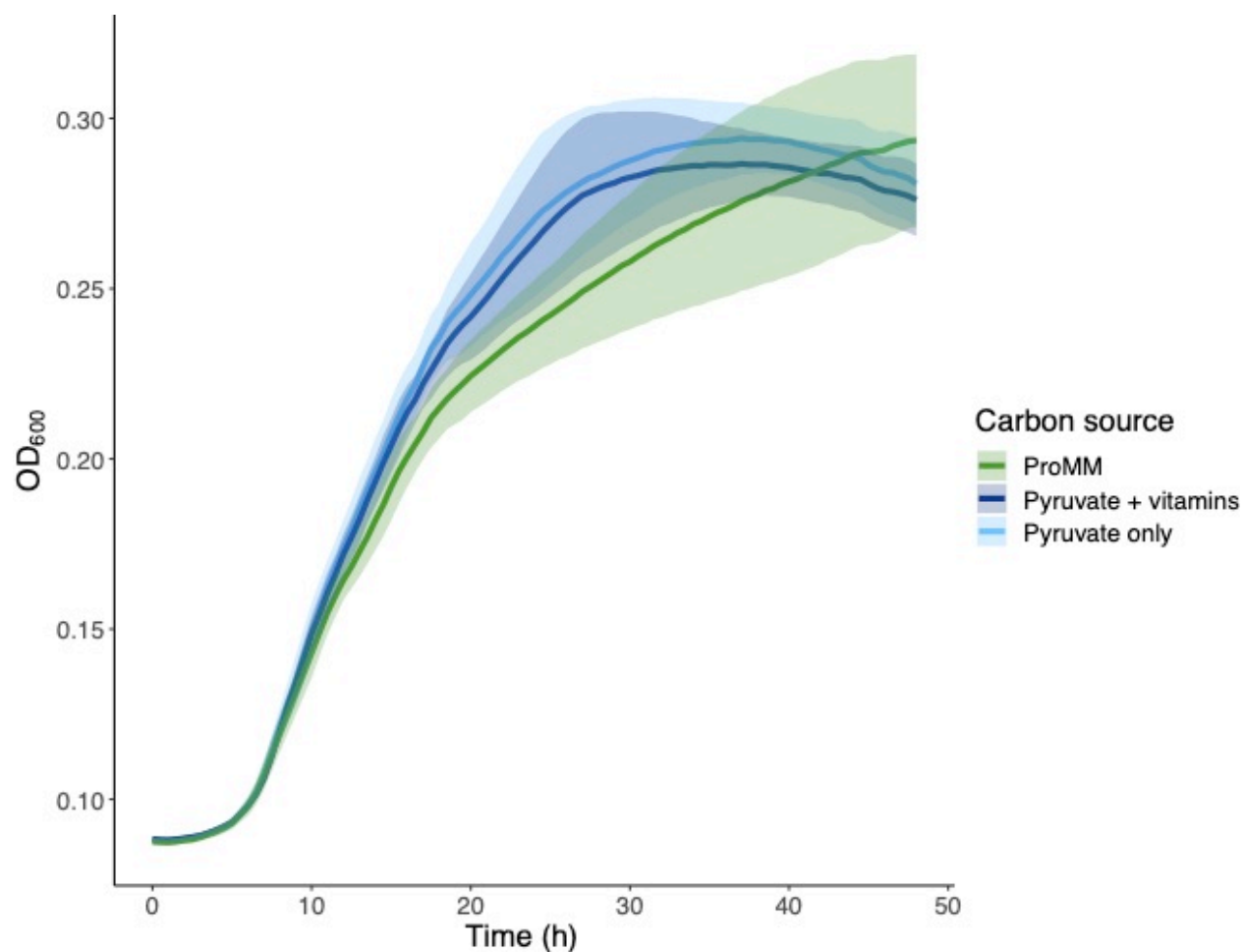
82 MIT1002 genes, and their similarity (if found) to *A. macleodii* MIT1002 genes.

Gene	Organism	Reference	UniProtKB accession #	NCBI reference	E-value	% identity
DctM	<i>Rhodobacter capsulatus</i> SB 1003	Thomas et al. 2006	D5ATK1	WP_050759969.1	NA	NA
DctQ	<i>Rhodobacter capsulatus</i> SB 1003	Thomas et al. 2006	D5ATK0	WP_013067268.1	NA	NA
DctP	<i>Rhodobacter capsulatus</i> SB 1003	Thomas et al. 2006	D5ALT6	WP_013068126.1	NA	NA
Alr3026	<i>Anabaena (nostoc)</i> PCC7120	Pernil et al. 2010	Q8YSQ8	NA	NA	NA
Alr3027	<i>Anabaena (nostoc)</i> PCC7120	Pernil et al. 2010	Q8YSQ7	NA	5e-22	30.33%
All3028	<i>Anabaena (nostoc)</i> PCC7120	Pernil et al. 2010	Q8YSQ6	NA	NA	NA

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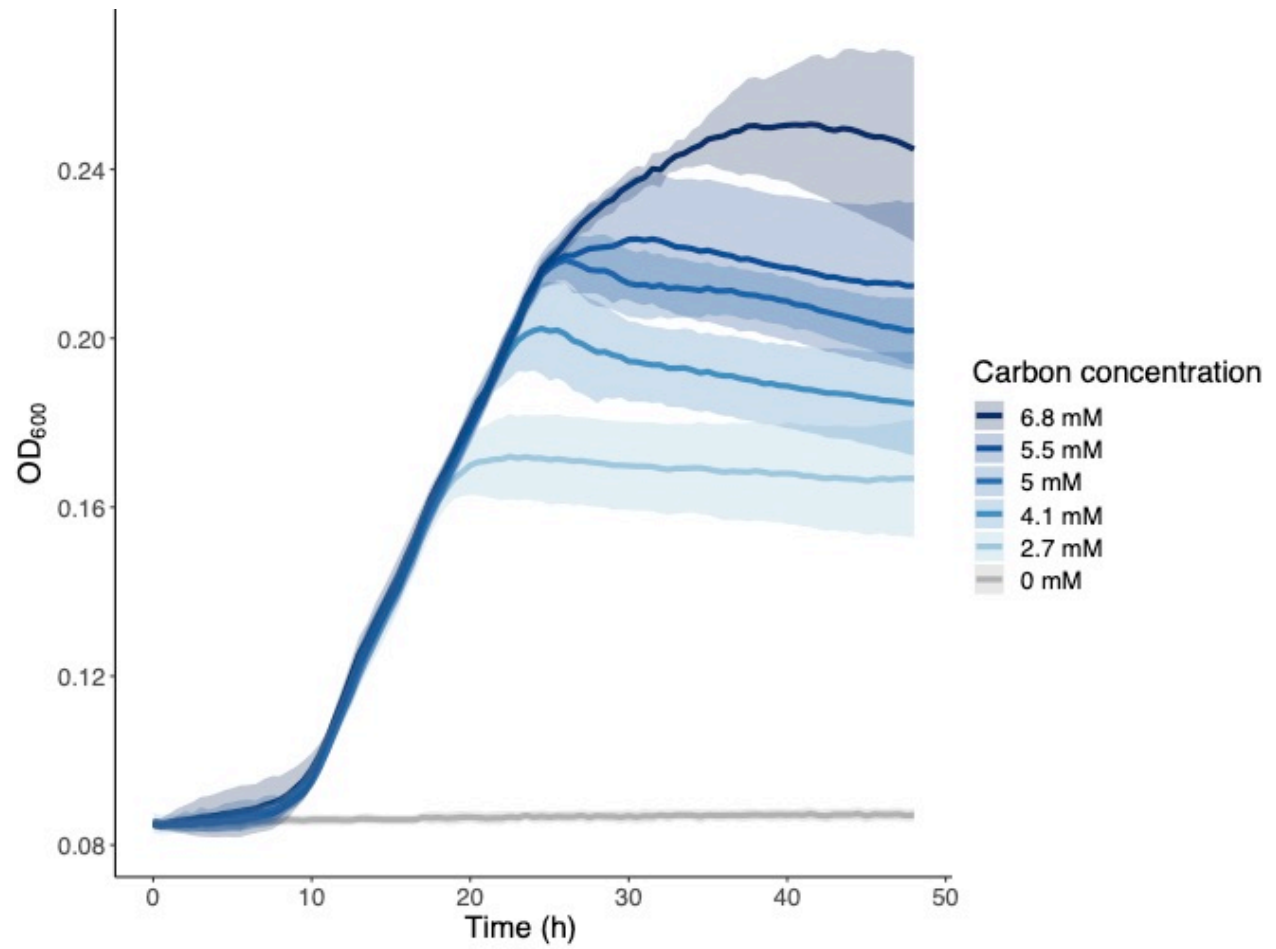
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85 Supplemental Figures



86

87 Figure S1. *Alteromonas* growth over 48 hours on either ProMM media, pyruvate (6.9 mM
 88 carbon, matching total ProMM carbon concentration) and vitamins (Va vitamin mix, matching
 89 the ProMM vitamin addition), or pyruvate alone (6.9 mM carbon). Carrying capacity was not
 90 significantly different between conditions (ANOVA, $p > 0.05$).



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92 Figure S2. *Alteromonas* growth on varying concentrations of carbon from pyruvate, from 6.8

93 mM to 2.7 mM carbon.

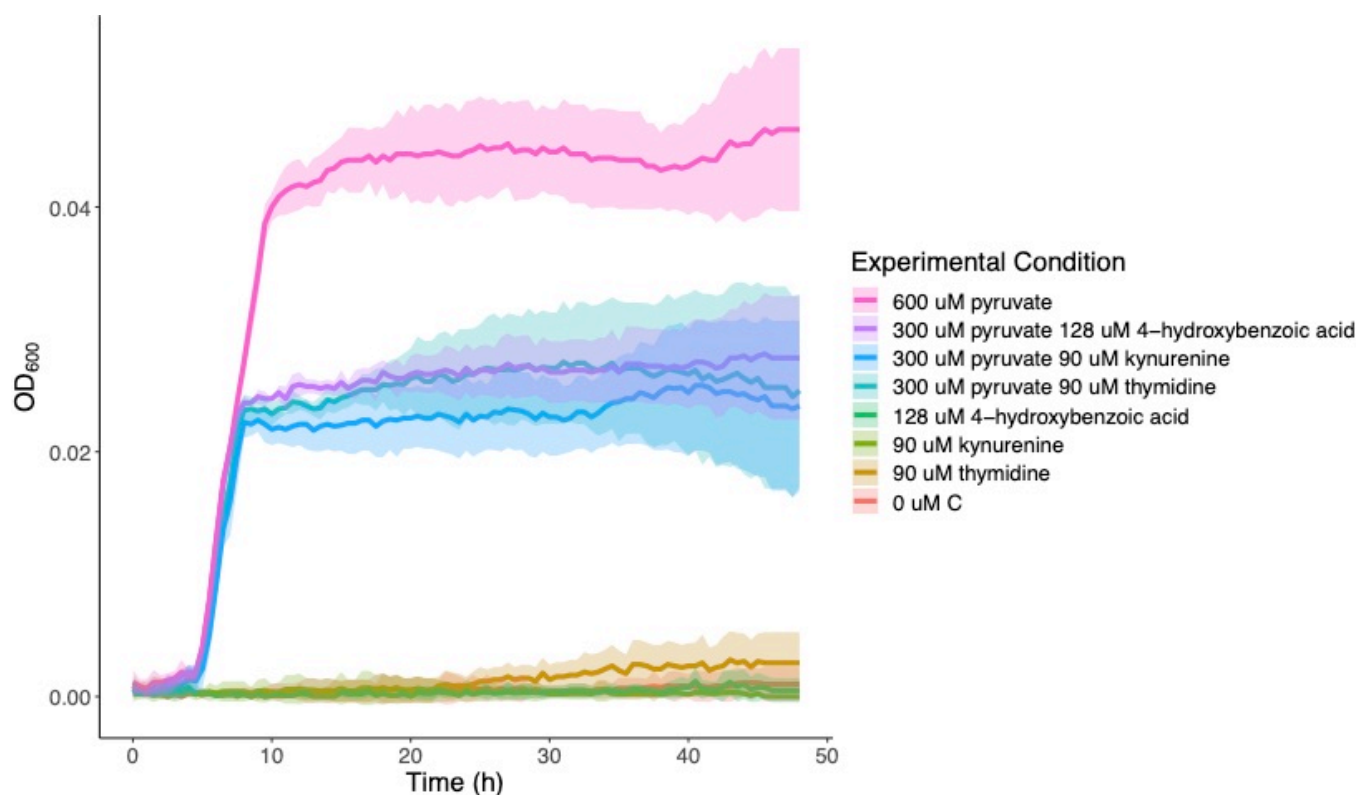


Figure S3. Growth curves of *Alteromonas* incubated with metabolites which cannot support growth. *Alteromonas* was grown with either 600 μ M pyruvate as a positive control, no added carbon as a negative control, an example metabolite (4-hydroxybenzoic acid, kynurenine, or thymidine) which were could not support growth in initial experiments, or a mix of pyruvate and an example metabolite. Total carbon concentrations in the mixed substrate cultures are equivalent to the carbon received from 600 μ M pyruvate (1.8 mM).

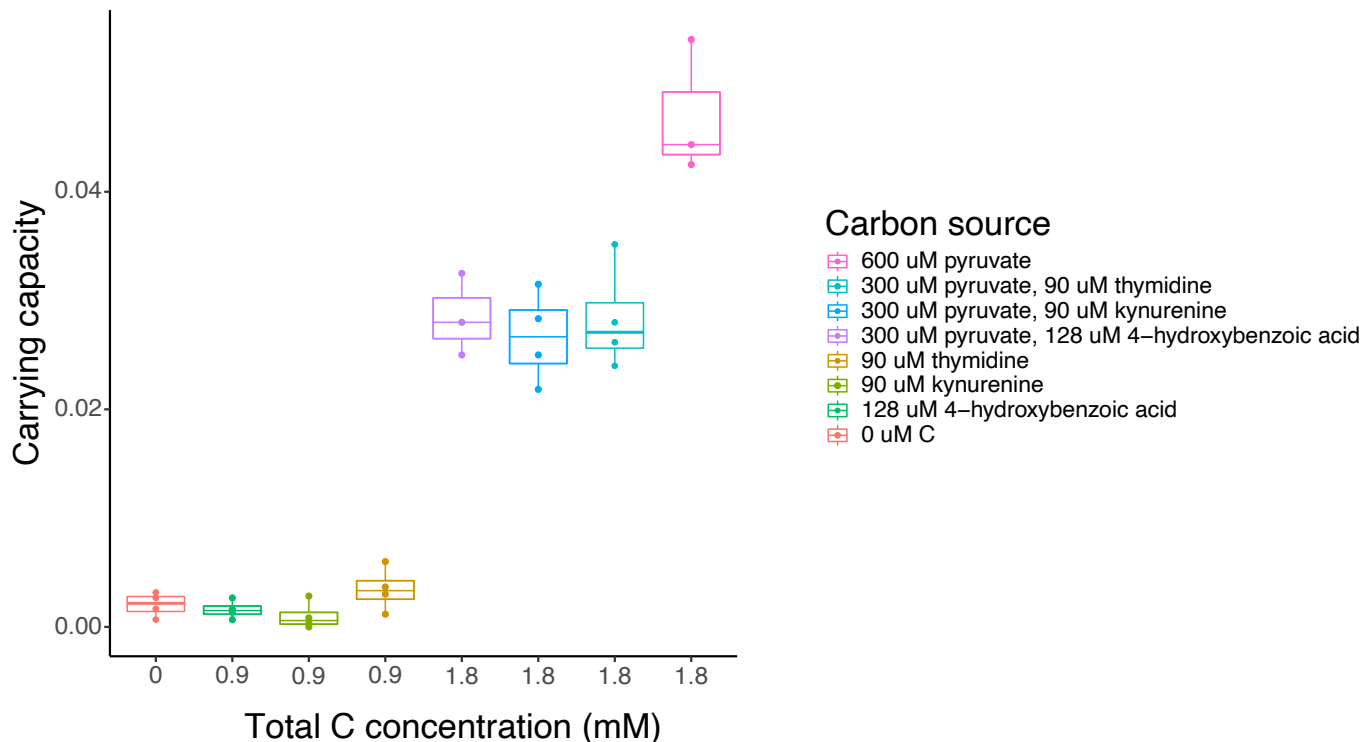


Figure S4. Calculated carrying capacities of *Alteromonas* cultures incubated with pyruvate, a *Prochlorococcus* metabolite which cannot support growth, or a mix of pyruvate and one of these metabolites. There was no significant difference in carrying capacity between cultures which received 0 mM carbon and those which received 0.9 mM C from any of the three carbon sources, and no significant difference in carrying capacity between cultures which received 1.8 mM carbon from a mix of pyruvate and any one metabolite (two way ANOVA with post-hoc Tukey's HSD test, $p > 0.05$). *Alteromonas* grown on 1.8 mM C from pyruvate reached significantly higher carrying capacities than *Alteromonas* growth with 1.8 mM C from a mix of pyruvate and a *Prochlorococcus* metabolite (two way ANOVA with post-hoc Tukey's HSD test, $p < 0.05$).

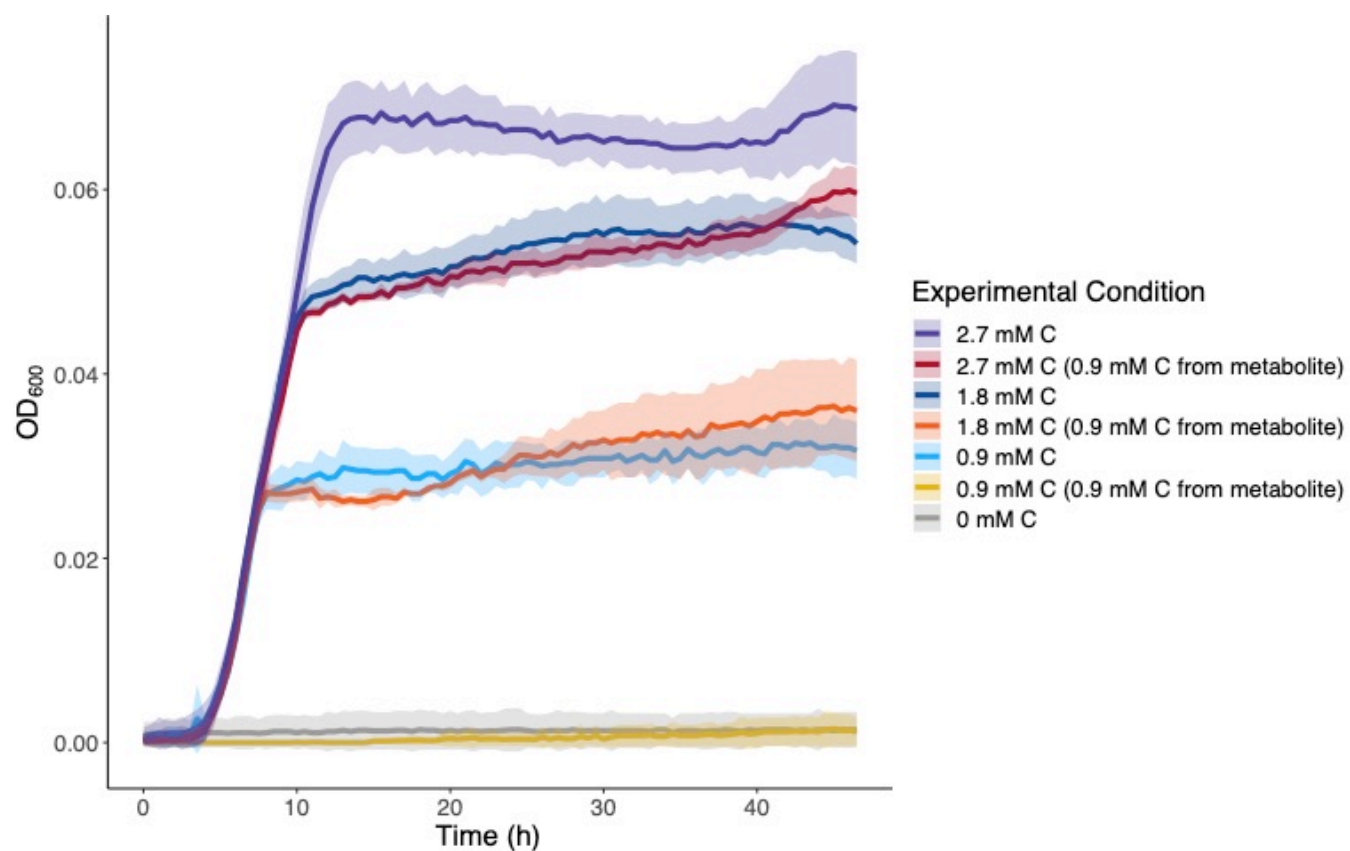
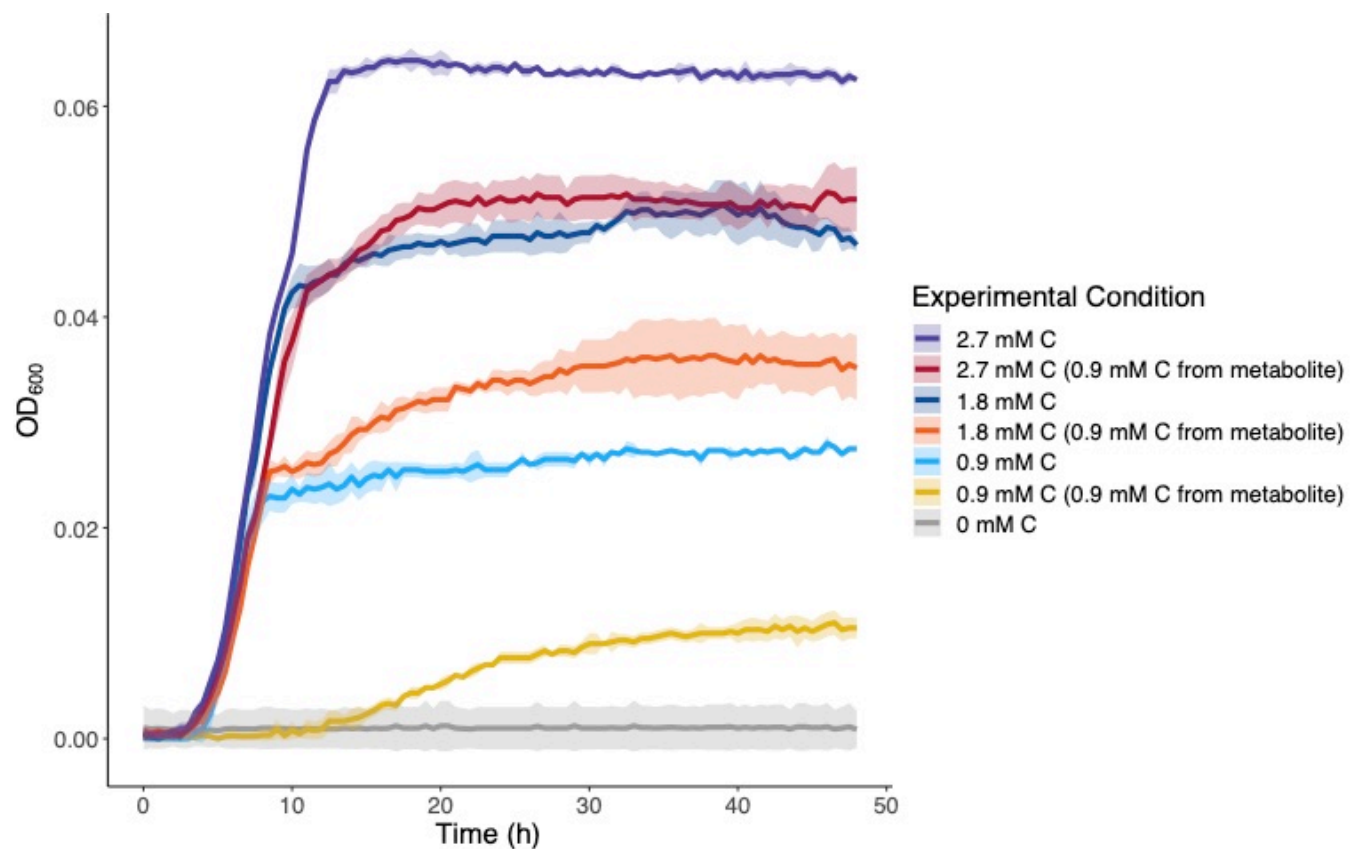


Figure S5. *Alteromonas* growth on 4-methyl-2-oxopentanoic acid, pyruvate, or a mix of 4-methyl-2-oxopentanoic acid and pyruvate, across a range of carbon concentrations. Plot shows mean OD₆₀₀ $\pm$ 1 SD (N $\geq$ 3).

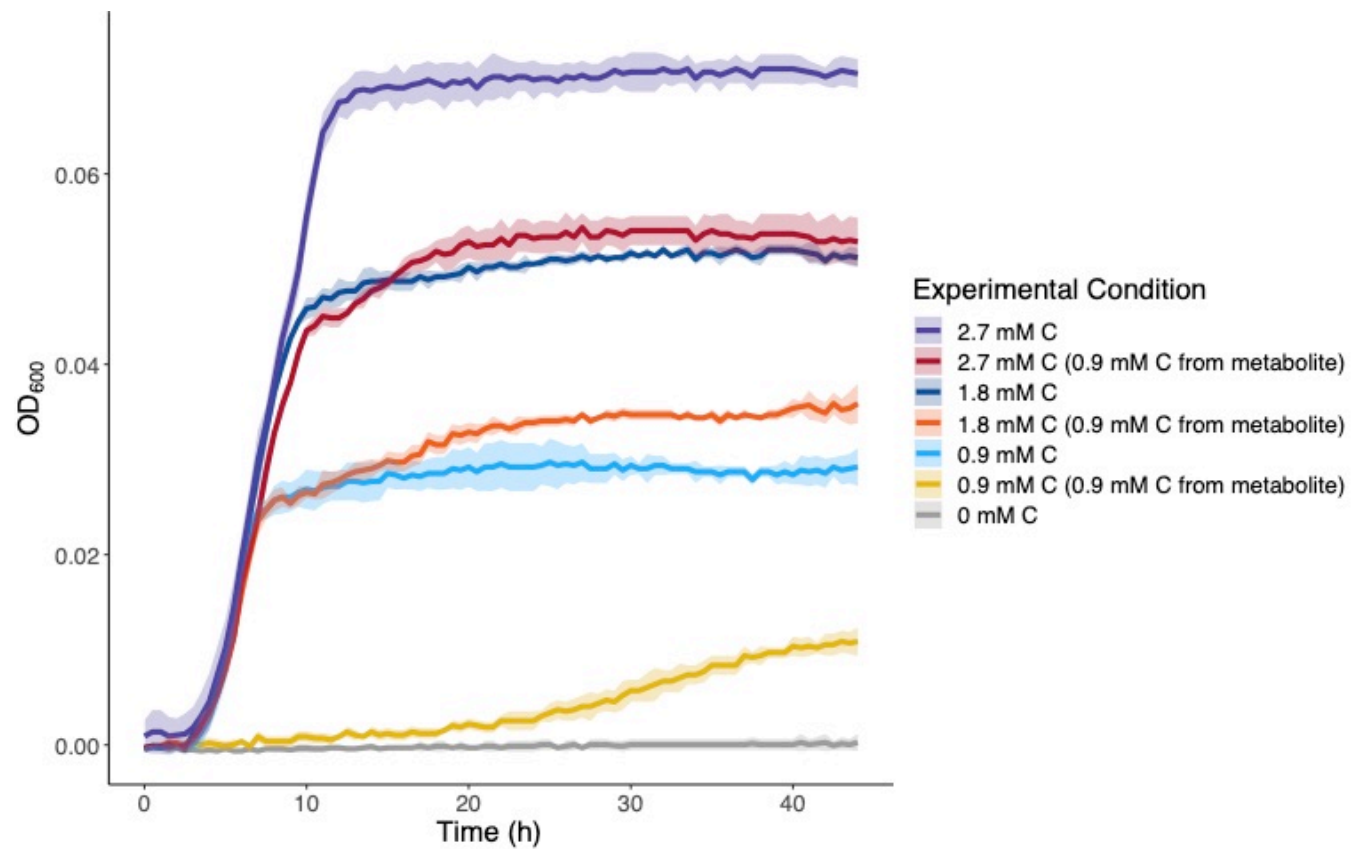


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119 Figure S6. *Alteromonas* growth on leucine, pyruvate, or a mix of leucine and pyruvate, across a

120 range of carbon concentrations. Plot shows mean OD₆₀₀ ± 1 SD (N ≥ 3, except for the 900 μM

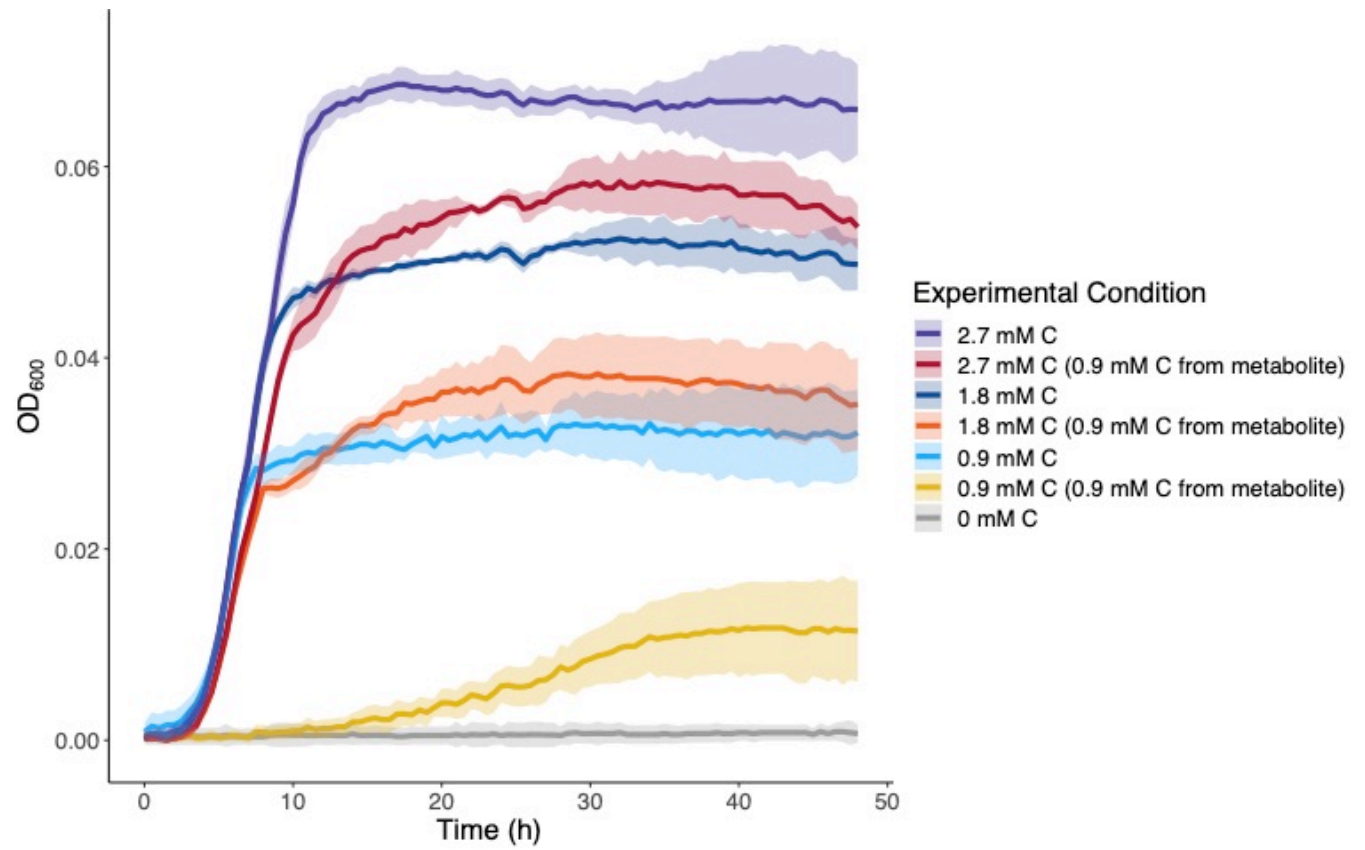
121 C from pyruvate condition, where N = 2 due to the removal of edge rows or columns).



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123 Figure S7. *Alteromonas* growth on isoleucine, pyruvate, or a mix of isoleucine and pyruvate,

124 across a range of carbon concentrations. Plot shows mean OD₆₀₀ $\pm$ 1 SD (N $\geq$ 3).



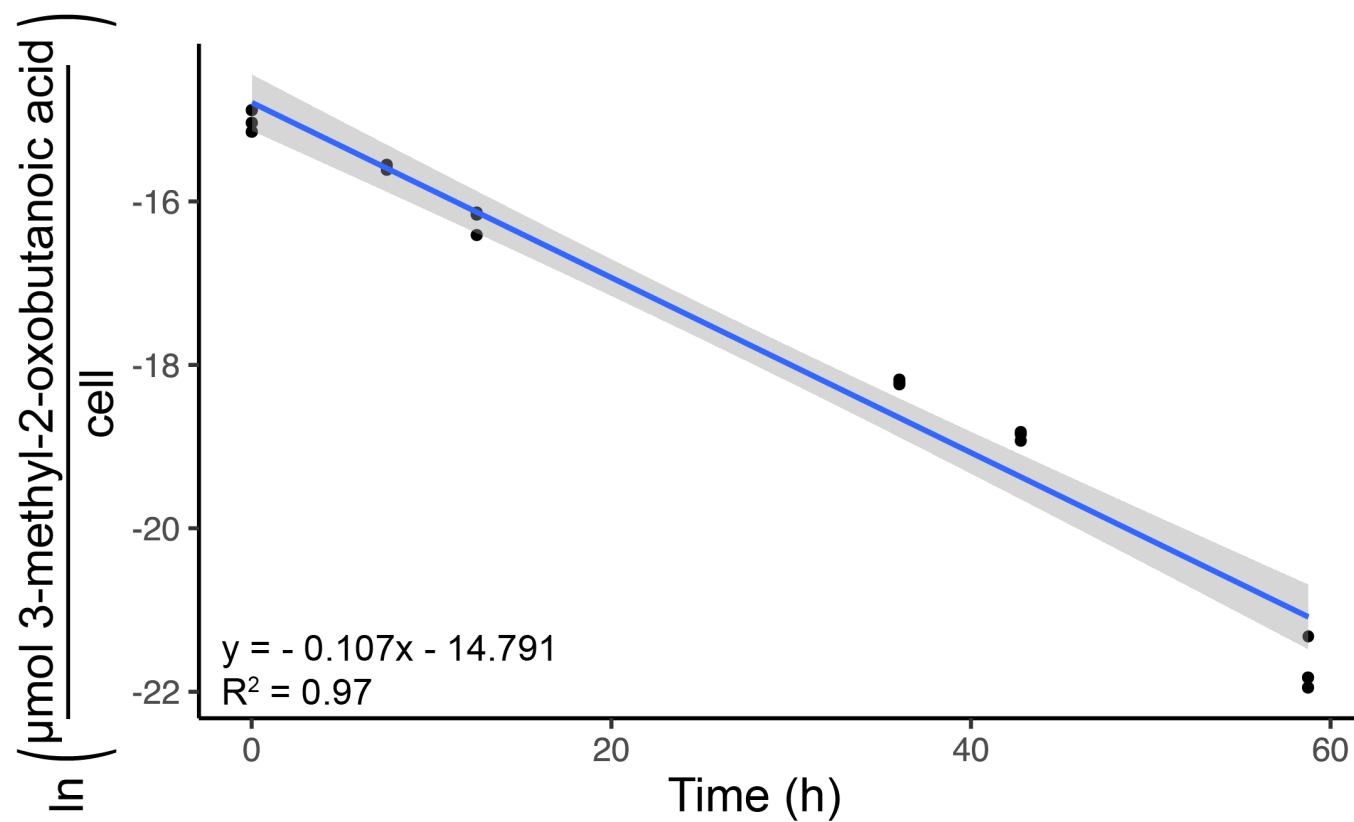
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Figure S8. *Alteromonas* growth on valine, pyruvate, or a mix of valine and pyruvate, across a

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range of carbon concentrations. Plot shows mean OD₆₀₀ $\pm$ 1 SD (N $\geq$ 3).



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130 Figure S9. Cell-normalized concentrations of extracellular 3-methyl-2-oxobutanoic acid over the
131 course of a 59 hour incubation with *Alteromonas macleodii* MIT1002.

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133 **Supplemental Works Cited**

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